

Day 3:

Dictionary - {}

Student = {

    "name" : "navoeh",

    "age" : 21,  
}

student = {

output:

```
{ "name": "naveen", "age": 21 }
```

type(student)

dict

key call → value  
name call → student

```
Fruit = {  
    "apple": 100  
    "Banana": 40  
}  
print(Fruit["banana"])
```

output  
40

```
Fruit["banana"] = 28  
print(Fruit)
```

output:

```
{ 'apple': 100, 'banana': 28 }
```

Fruit.pop("apple") → apple delete  
100 no delete

In Function dict:

```
def student_details():  
    student = {  
        "name" = "Arthi",  
        "marks" = "100"  
    }  
    print(student["name"])  
    print(student["marks"])  
student_details()
```

output:

Arthi  
100



## Loops:

\* A loop is a programming structure that repeats a block of code until a condition becomes false until all items are processed

### Why we use loop:

- ⇒ Code repetition X
- ⇒ large Amount of data handle
- ⇒ Repeated code Automation.

### 2 loops:

- \* For loop (all items to be processed)
- \* While loop (

### For loop:

When we know how many times we want to execute or repeat something.

### Syntax:

For Variable in Sequence → range (or) continuous value  
Statement → what process

ex:

For i in range(5) → Build in Function  
Print(i) → How many times.

### Output:

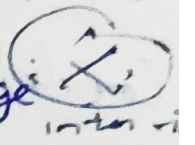
0  
1  
2  
3  
4

For i in range (1, 6) → exclusive  
print(i) → inclusive  
output:

1  
2  
3  
4  
5

Range (2, 20, 3) → step size  
output:

2  
5  
8  
11  
14  
17

3 ~~type~~ Range 

1st ⇒ How many time run  
2nd ⇒ 1 to 6  
3rd ⇒ step size

loop through a list:

Fruits = ["apple", "banana", "orange"]  
For fruit in Fruits:  
print (Fruits)

output:

apple  
banana  
orange

loop through a string:

name = "python"  
For letter in name:  
print (letter)

output:

p  
y  
t  
h  
o  
n



while loop:  
\* Statement

Condition

Must be

True (or) False

\* While loops Runs a longest the Condition is true on the Condition. False, Depends Upon our Condition.

Syntax:

while Condition :  
Statement

ex:

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```

increase by 1

Output:

1  
2  
3  
4  
5

ex:

```
while i <= 5:  
    print(i)  
    i = i + 1
```

output:  
6

(Wrong logic)  
(Random Belect)

Nested loop:

\* loop inside loop

ex:

```
For i in range(3):  
    For j in range(2):  
        print(i, j)
```

Output:

0 0  
0 1  
1 0  
1 1  
2 0  
2 1

# loop Control:

Break

Continue

Pass

→ Build in function

break:

For i in range(10):

if i == 5 → break

~~break~~

print(i)

output:

0  
1  
2  
3  
4

Continue:

For i in range(5):

if i == 2 → skip current iteration and  
continue

print(i)

output:

0  
1  
3  
4

Infinite loop:

while True:

print("hello world")

output

hello world run

Many time.

Ex:

op = [1, 3, 5, 7, 9]

For a in op:

if a % 5 == 0

print(F"(a) is divisible by 5")

else

print(F"(a) is not divisible by 5")



Output:

1 is not divisible by 5  
3 is not " "  
5 is divisible by 5  
7 is not divisible by 5  
9 is not divisible by 5

lup = (1, 7, 9, 6, 4, 2)

For lup:

if i % 2 == 0:  
print(f"{i} is a even number")

output:

6 is a even number  
4 " "  
2 " "

loop through list:

Fruits = ["apple", "banana"]

For i in Fruits:

print("I like ", i)

output:

I like apple  
I like banana.

ATM pin

attempts = 0

pin\_ok = false

while attempts < 3:

pin = input("enter")

if pin == "1234"

pin\_ok = True

break

attempts + 1

else:

print("too many attempts")

if pin\_ok == True:

print("Access granted")

2n:  
\* 3 tym wrong password loop exit otherwise  
login?

Concise:

```
str1 = "Hello"  
str2 = "world"  
result = str1 + str2  
print(result)
```

(or)

```
str1 = "Hello"  
str2 = "world"  
result = "".join(str1, str2)  
print(result)
```

```
name = "John"
```

```
age = 20
```

```
print(f"my name is {name}, I am {age}")
```

output:

my name is John, I am 20